

DAY 9 MYSQL ASSINGNMENT

```
USE slatask;
```

```
DROP TABLE Products;
```

```
CREATE TABLE Products(  
product_id INT PRIMARY KEY AUTO_INCREMENT,  
name VARCHAR(100),  
price DECIMAL(10,2),  
stock_quantity INT,  
status VARCHAR(20) DEFAULT 'Available'  
);
```

```
CREATE TABLE Inventory_Audit(  
audit_id INT PRIMARY KEY AUTO_INCREMENT,  
product_id INT,  
action_type VARCHAR(50),  
old_value VARCHAR(100),  
new_value VARCHAR(100),  
changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
INSERT INTO Products(name,price,stock_quantity)
```

```
VALUES
```

```
('Mouse',500,10),
```

```
('Keyboard',800,5),
```

```
('Monitor',7000,2);
```

```
SELECT * FROM Products;
```

	product_id	name	price	stock_quantity	status
▶	1	Mouse	500.00	10	Available
	2	Keyboard	800.00	5	Available
	3	Monitor	7000.00	2	Available
*	NULL	NULL	NULL	NULL	NULL

```
-- 1 Prevent Negative Stock
```

```
DELIMITER //

CREATE TRIGGER t1
BEFORE UPDATE ON Products
FOR EACH ROW
BEGIN
IF NEW.stock_quantity < 0 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT='Negative stock not allowed';
END IF;
END //

DELIMITER ;
```

-- 2 Audit Price Change

```
DELIMITER //

CREATE TRIGGER t2
AFTER UPDATE ON Products
FOR EACH ROW
BEGIN
IF OLD.price <> NEW.price THEN
INSERT INTO Inventory_Audit(product_id,action_type,old_value,new_value)
VALUES(OLD.product_id,'Price Change',OLD.price,NEW.price);
END IF;
END //

DELIMITER ;
```

-- 3 Auto Out of Stock

```
DELIMITER //

CREATE TRIGGER t3
BEFORE UPDATE ON Products
FOR EACH ROW
BEGIN
```

```
IF NEW.stock_quantity=0 THEN
SET NEW.status='Out of Stock';
END IF;
END //
DELIMITER ;
```

-- 4 Uppercase Name

```
DELIMITER //
CREATE TRIGGER t4
BEFORE INSERT ON Products
FOR EACH ROW
BEGIN
SET NEW.name=UPPER(NEW.name);
END //
DELIMITER ;
```

-- 5 Log New Product

```
DELIMITER //
CREATE TRIGGER t5
AFTER INSERT ON Products
FOR EACH ROW
BEGIN
INSERT INTO Inventory_Audit(product_id,action_type,new_value)
VALUES(NEW.product_id,'New Product',NEW.name);
END //
DELIMITER ;
```

-- 6 Prevent Delete Active Product

```
DELIMITER //
CREATE TRIGGER t6
```

```
BEFORE DELETE ON Products
FOR EACH ROW
BEGIN
IF OLD.stock_quantity>0 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT='Cannot delete active product';
END IF;
END //
DELIMITER ;
```

```
-- 7 Track Deleted Product
DELIMITER //
CREATE TRIGGER t7
AFTER DELETE ON Products
FOR EACH ROW
BEGIN
INSERT INTO Inventory_Audit(product_id,action_type,old_value)
VALUES(OLD.product_id,'Deleted',OLD.name);
END //
DELIMITER ;
```

```
-- 8 Minimum Price
DELIMITER //
CREATE TRIGGER t8
BEFORE INSERT ON Products
FOR EACH ROW
BEGIN
IF NEW.price<1 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT='Price must be >=1';
END IF;
```

END //

DELIMITER ;

-- 9 Stock Increase Log

DELIMITER //

CREATE TRIGGER t9

AFTER UPDATE ON Products

FOR EACH ROW

BEGIN

IF NEW.stock_quantity>OLD.stock_quantity THEN

INSERT INTO Inventory_Audit(product_id,action_type)

VALUES(NEW.product_id,'Stock Added');

END IF;

END //

DELIMITER ;

-- 10 Prevent Weekend Price Change

DELIMITER //

CREATE TRIGGER t10

BEFORE UPDATE ON Products

FOR EACH ROW

BEGIN

IF DAYOFWEEK(CURDATE()) IN(1,7)

AND OLD.price<>NEW.price THEN

SIGNAL SQLSTATE '45000'

SET MESSAGE_TEXT='Weekend price change not allowed';

END IF;

END //

DELIMITER ;